SHEAFIFYING THE PRESHEAF OF COHOMOLOGY GROUPS OF AN OBJECT RECOVERS ITS COHOMOLOGY SHEAF

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument. The author continues the sentence with a lower-case 'by our construction of cohomology' after the displayed formula (source line 4368); that capitalization is reproduced verbatim and no wording was normalised.

Editorial note. Cross-chapter restatement: the same label occurs in chapter cohomology for a ringed space. That statement is the special case of this ringed-site statement, so per the lane-s4 rule only this more general occurrence is counted here.

Editorial note. The single labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer to the same chapter.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Cohomology sheaf H^q(K) of an object of D(O). Same chapter, the definition of the cohomology sheaves of a complex of O-modules. Supplies the object whose identification with a sheafification is asserted.
- Cohomology H^q(U, K) computed from a K-injective representative. Same chapter, the computation of H^q(U, K) as kernel modulo image in the section complex of a K-injective representative. Supplies the explicit formula for the presheaf being sheafified.
- Sheafification of a presheaf of modules on a site. Same chapter, the sheafification of a presheaf of O-modules and its universal property. Supplies the operation applied to the presheaf U -> H^q(U, K) in the statement.

Source reference. Original lemma 0D6F: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 4329–4329. Same-chapter navigation pointer invoked at proof line 4360 for the identification of H^q(U, K) with H^q(C/U, K|_{C/U}). The pointer only resolves that reference for the reader; the cited lemma is displayed again by the author inside the counted proof, so nothing load-bearing is replaced, and no bibliography citation is involved.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let K be an object of $D(\mathcal{O})$. The sheafification of*

$$U \mapsto H^q(U, K) = H^q(\mathcal{C}/U, K|_{\mathcal{C}/U})$$

is the qth cohomology sheaf $H^q(K)$ of K.

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Proof. The equality $H^q(U, K) = H^q(\mathcal{C}/U, K|_{\mathcal{C}/U})$ holds by Lemma 0D6F. Choose a K-injective complex $\mathcal{I}^\bullet$ representing K. Then

$$H^q(U, K) = \frac{\mathrm{Ker}(\mathcal{I}^q(U) \to \mathcal{I}^{q+1}(U))}{\mathrm{Im}(\mathcal{I}^{q-1}(U) \to \mathcal{I}^q(U))}.$$

by our construction of cohomology. Since $H^q(K) = \mathrm{Ker}(\mathcal{I}^q \to \mathcal{I}^{q+1})/\mathrm{Im}(\mathcal{I}^{q-1} \to \mathcal{I}^q)$ the result is clear. □